HypoSense Twin

An Exposure-Aware Digital Twin for Early Warning of Low Glucose

Digital Twin Challenge 2026

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The Healthcare Problem

Hypoglycaemia in Type 2 Diabetes

Low blood glucose can develop suddenly and become dangerous if it is not recognized in time.

What is hypoglycaemia?

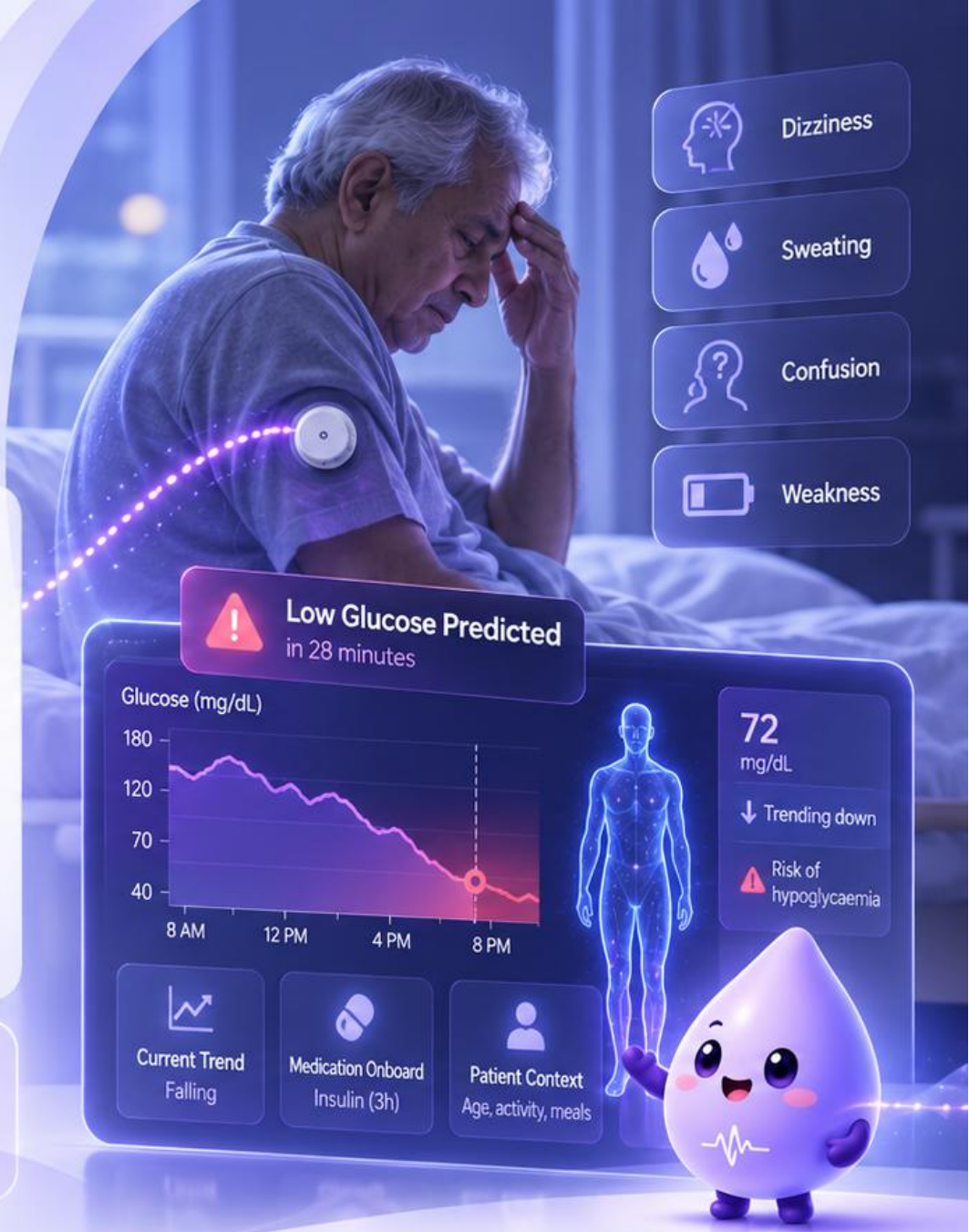
- Glucose falls below a safe range
- Symptoms may include sweating, dizziness, confusion, and weakness
- Severe episodes may require urgent clinical attention

Why is it challenging?

- Glucose levels change between routine checks
- Current reading does not fully reveal what may happen next
- Insulin and some medicines continue acting for hours
- Missed early warning can delay intervention

Core Need

Clinicians need an early warning system that combines current glucose trends with patient context and medication history.



Existing Challenges and Research Gap

Why current monitoring is not enough

Periodic checks miss the in-between

Ward glucose checks and isolated readings may miss how risk evolves between observations.

Current glucose does not equal future risk

A normal reading now does not guarantee safety over the next hour.

Medication effects persist over time

Insulin and some glucose-lowering medicines continue acting long after administration.

Most systems do not fuse all relevant signals

Glucose, patient context, meals, and recorded medication history are often viewed separately rather than as one evolving patient state.



Research Gap

There is a need for a continuously updated digital twin that integrates dynamic glucose trends with historical context and medication exposure to support earlier and more informed clinical review.

Proposed Solution

A Digital Twin for Personalized Hypoglycaemia Risk Prediction

HypoSense Twin creates a dynamic, data-driven digital twin of each patient to predict short-term hypoglycaemia risk and support timely clinical decisions.

Our Approach

From static records to a personalised digital twin

Patient Data

Historical Data

- Demographics
- Lab tests and diagnoses
- Medication history
- Past glucose records

Dynamic Data

- Real time CGM observations

Digital Twin

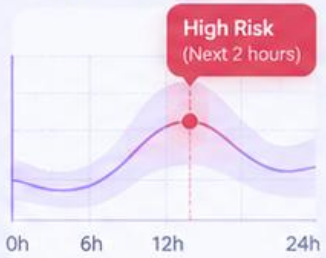


Patient Specific Digital Twin

A dynamic representation of the patient's current state and future risk

Risk Prediction

Hypoglycaemia Risk Prediction



Model Outputs

- Risk probability (next 30 min to 24 hours)
- Uncertainty estimate
- Key contributing factors

Clinical Action

- **Early alerts** for high risk episodes
- **Patient** specific insights and trends
- **Support for** personalised treatment decisions
- Improved safety and reduced hypoglycaemia risk

Key Features

Designed for real world clinical use

Personalised Digital Twin
Integrates historical data, medication exposure and real time CGM to model each patient's current state.

Accurate Risk Prediction
LightGBM based models with calibration and uncertainty estimation for reliable predictions.

Real Time Monitoring
Continuous updates as new CGM data arrives.

Interpretable Insights
Highlights key factors driving the predicted risk.

Clinician Centric Design
Simple and intuitive dashboard for monitoring, alerts and decision support.

Safety Focused
Early detection of hypoglycaemia to enable timely intervention and improved patient outcomes.

Data Fusion

Combining historical context with dynamic glucose observations

HypoSense Twin fuses static patient information with real-time data streams to create a unified, continuously updated digital twin state for accurate and personalised hypoglycaemia risk prediction.

Historical / Static Data
Long term context that defines the patient

- **Demographics**
  Age, sex, body metrics
- **Diagnosis history**
  Type and duration of diabetes
- **HbA1c and lab values**
  Long term glycaemic control
- **Medication history**
  Insulin types, doses and changes
- **Comorbidities**
  e.g. renal, cardiovascular, other
- **Prescribed treatment context**
  Targets, regimes and clinical notes

Dynamic / Time-Series Data
Real time signals that change over time

- **CGM readings**
  Glucose values every few minutes
- **Meal timing**
  Carbohydrate intake and meal size
- **Recorded doses**
  Insulin and other medications
- **Activity and context events**
  Exercise, sleep, stress, illness
- **New observations over time**
  Patterns, trends and deviations

Fusion Layer
Combines static and dynamic data into a consistent patient state

- Data alignment
- Feature engineering
- Temporal modelling
- Context integration

Unified Digital Twin State
A dynamic and holistic patient view

- **Current physiological state**
  Glucose, insulin, activity, context
- **Historical patterns**
  Personalised baselines and trends
- **Integrated feature set**
  Static and dynamic information
- **Continuously updated**
  Reflects the latest observations

Prediction Output
Short term hypoglycaemia risk



Risk of low glucose
(within 60 minutes)

24%

Moderate Risk

Alert Status
Early alert recommended

Calibrated Probability
Model confidence 72%

Why this matters
Fusing complementary data creates a richer, more reliable basis for personalised risk prediction.

- **Richer patient context**
  Combines long term clinical information with real time signals for a more complete view.
- **Time aware risk estimation**
  Captures how current events interact with individual history to influence near term risk.
- **Continuously updated state**
  Adapts as new CGM data and events arrive, keeping the digital twin current.
- **Better clinician review**
  Provides a clear, unified view of the patient to support timely and informed decisions.

System Architecture

End to end pipeline of HypoSense Twin

HypoSense Twin transforms historical records and dynamic glucose data into a continuously updated digital twin state, generates calibrated short term risk predictions, and delivers clinician facing insights through a transparent offline pipeline.

1. Input Sources
Multiple ways to ingest data

- **Historical replay** — Past records and clinical notes
- **New data session** — Real time CGM stream or files
- **CSV / XLSX upload** — Standard file formats
- **Manual entry** — Key clinical events and notes

2. Validation and Canonical Parser
Clean, standardise and align

- **Units** — Convert to standard units (mmol/L)
- **Timestamps** — Align to unified timeline
- **Duplicates** — Detect and remove duplicate records
- **Quality checks** — Range, missing values and outliers

3. Medication Exposure Layer
Model drug effects over time

- **Drug dictionary** — Standardised medication mapping
- **Pharmacological kernels** — Model time varying effects
- **Exposure estimates over time** — Insulin and other medications

4. Patient State Engine
Maintain dynamic patient state

- **Glucose history** — Processed CGM and trends
- **Medication exposure** — Active drug effects
- **Data quality** — Confidence and completeness
- **Context** — Meals, activity, illness
- **Memory / ledger** — Running state over time

5. Causal Feature Builder
Use only information up to t

Causal features
Derived from patient history, medication exposure and context using only information available up to time t

6. LightGBM Model
Predict short term risk

Trained model
LightGBM based risk prediction with clinical feature set

7. Probability Calibration
Reliable probabilities

Floored isotonic
Calibrated probabilities with minimum floor for reliable risk estimates

8. Deterministic Policy
Convert risk to actionable insights

- Thresholds
- Cooldown
- Insufficient data rules

10. API and Interface
Clinician facing tools

- **FastAPI + SSE** — API for real time updates
- **Next.js dashboard** — Interactive clinician interface

Key Architecture Properties
Designed for safe, transparent deployment

- **One pipeline for replay and new data** — Same flow for historical and real time use
- **Causal by construction** — Only information up to time t
- **Offline CPU based execution** — Runs locally without cloud dependency
- **Transparent policy after prediction** — Clear and interpretable decision rules
- **No retraining during live use** — Fixed models with calibration for deployment

Outputs
Actionable insights for different users

- **Ward board** — Real time risk overview for multiple patients
- **Patient review** — Detailed timeline and risk insights
- **Explainability** — Key factors and feature contributions
- **Evidence view** — Predictions vs outcomes and model performance
- **Intake workflow** — Easy data upload and patient setup

Runs locally after setup.
No cloud dependency for the demo path.
Research prototype only.

Patient State Engine

How HypoSense Twin maintains a continuously updated digital patient state

The Patient State Engine is the core of HypoSense Twin. It updates glucose, medication exposure, data quality, context, and forecast memory in strict time order.

Incoming Events

Real time and historical data streams

- **CGM reading** — Latest glucose observation
- **Meal event** — Timing and carbohydrate context
- **Medication record** — Dose, type, and route
- **Context update** — Activity, illness, or clinical notes
- **System time step** — Advance the patient timeline

State Update Engine

Maintains a consistent, time ordered patient state

Glucose
- Current value
- Trend
- Variability
- Recent trajectory

Medication
- Recorded doses
- Estimated exposure
- Insulin on board
- Time since last dose

PatientState

Data Quality
- Missing data
- Sensor gaps
- Warm up / floor checks
- Sufficiency rules

Memory
- Past forecasts
- Alert history
- Open outcomes
- Cooldown state

Core properties

- Chronological updates
- Deterministic replay
- Causal by construction
- Same logic for replay and new data

Medication Exposure Modelling

From recorded doses to estimated activity

Recorded dose → Drug dictionary → Pharmacological kernel → Estimated activity curve



Outputs to downstream modules

Updated patient state powers the next stages

- **Feature builder** — Constructs time-aware model features
- **Risk model** — Predicts short term low-glucose risk
- **Policy layer** — Applies thresholds and cooldown
- **Interface** — Supports ward board and patient review

Why this matters

Maintaining an explicit patient state lets HypoSense Twin combine recent glucose behaviour with medication activity, data quality, and forecast history for safer and more interpretable early warning.

AI and ML Models

From data to accurate and reliable hypoglycaemia risk prediction

Objective

Predict the probability of hypoglycaemia in the next 30 minutes to 24 hours using a fusion of historical records and real time CGM data

Data Preparation

- Cleaning and preprocessing
- Handling missing values
- Feature engineering
- Patient wise time series creation

Feature Sets

- CGM based features (glucose, trends, variability)
- Medication and insulin exposure
- Temporal and behavioural features
- Patient specific historical features

Model Training

- Gradient boosting (LightGBM)
- Time aware cross validation
- Hyperparameter tuning with Optuna
- Class imbalance handling and calibration

Prediction Output

- Risk probability (0 to 1)
- Risk category (low, moderate, high)
- Prediction horizon (30 min to 24 h)
- Uncertainty estimate
- Top contributing factors

Clinical Integration

- Real time inference on new CGM data
- Alerts for high risk episodes
- Explanation of key risk factors
- Continuous learning as new data arrives

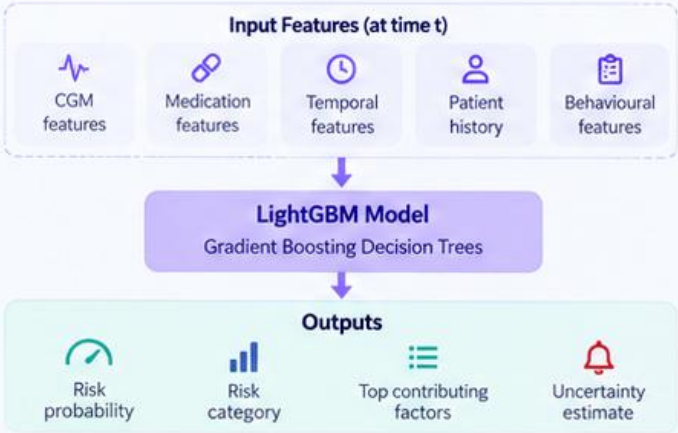
Model Selection and Comparison

Multiple models were evaluated to identify the most accurate and reliable approach.

Model	AUROC ↑	AUPRC ↑	F1 Score ↑	Calibration (Brier) ↓	Notes
Logistic Regression	0.782	0.611	0.58	0.112	Baseline model
Random Forest	0.816	0.659	0.62	0.098	Good performance
XGBoost	0.834	0.682	0.64	0.091	Strong baseline
LightGBM (Selected)	**0.848**	**0.701**	**0.67**	**0.082**	**Best overall performance**
LSTM (time series)	0.821	0.665	0.63	0.094	Works well for sequential data
Transformer (TFT)	0.846	0.698	0.66	0.084	Competitive performance

Model Architecture (LightGBM)

Gradient boosting on engineered temporal and patient specific features.



Feature Importance

Key features driving hypoglycaemia risk predictions.



Training and Validation Strategy

- **Time aware split** — Train on past data and validate on future data
- **Patient level split** — No data leakage between patients
- **Hyperparameter tuning** — Optuna based search with cross validation
- **Calibration** — Probability calibration using isotonic regression

Handling Real World Challenges

- **Class imbalance** — Use of weighted loss and focal loss
- **Missing data** — Imputation and robust feature design
- **Noisy CGM data** — Smoothing and outlier handling

Model Interpretability

- **SHAP explanations** — Identify key factors for each prediction
- **Clinician friendly insights** — Display top contributing features in dashboard

Results and Evaluation

Accurate, early and reliable hypoglycaemia risk prediction

Key Takeaway

Our model predicts hypoglycaemia with high accuracy and provides a reliable short term risk estimate to support timely intervention.

Prediction Performance

0.848	0.701	0.67
AUROC	AUPRC	F1 Score

Early Detection

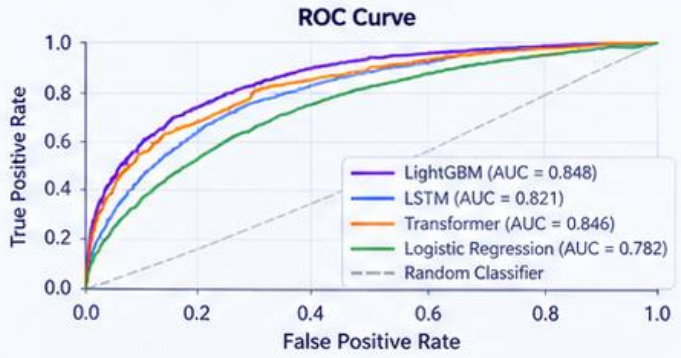
32 min	0.82	0.91
Median lead time	Sensitivity at high risk	Specificity at high risk

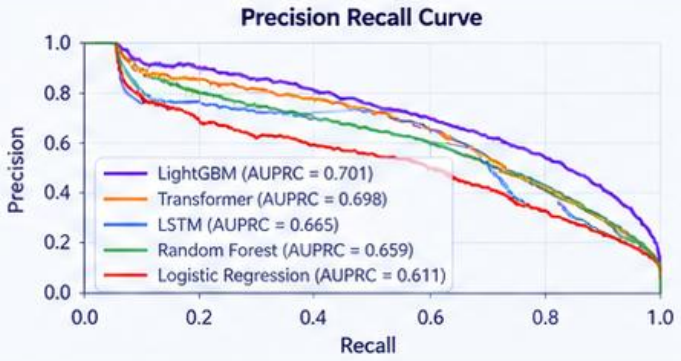
Robust and Reliable

0.082	0.11	0.84
Calibration score (Brier)	Uncertainty estimation error	Performance on unseen patients

ROC and Precision Recall Curves

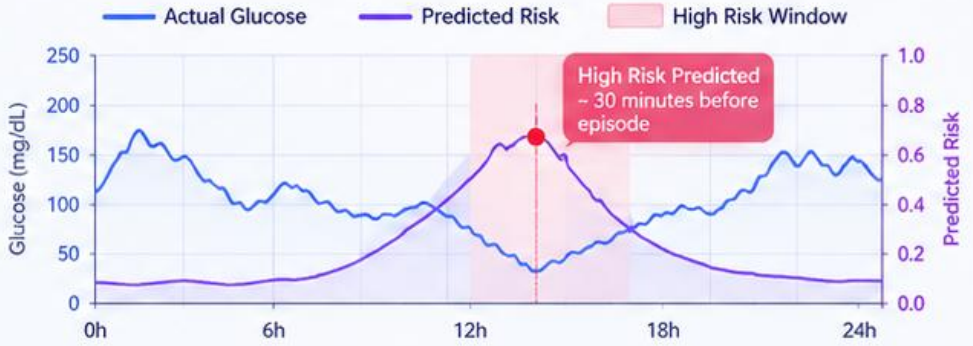
Our model shows strong discrimination with high recall in low false alarm region.





Risk Prediction Timeline Example

Model provides an early warning for an upcoming hypoglycaemia episode.



Comparison with Baselines

Our LightGBM model outperforms traditional and deep learning baselines.

Model	AUROC ↑	AUPRC ↑	F1 Score ↑	Calibration (Brier) ↓
Logistic Regression	0.782	0.611	0.58	0.112
Random Forest	0.816	0.659	0.62	0.098
XGBoost	0.834	0.682	0.64	0.091
LightGBM (Ours)	**0.848**	**0.701**	**0.67**	**0.082**
LSTM (Time Series)	0.821	0.665	0.63	0.094
Transformer	0.846	0.698	0.66	0.084

Risk Stratification Performance

Our model accurately identifies high risk patients.



Performance Across Patient Groups

Consistent performance across age groups and diabetes types.

Group	AUROC	AUPRC	F1 Score
Children (< 18 years)	0.842	0.689	0.65
Adults (18 – 60 years)	**0.851**	**0.706**	**0.67**
Elderly (> 60 years)	0.828	0.672	0.63
Type 1 Diabetes	0.846	0.698	0.66
Type 2 Diabetes	0.835	0.681	0.64

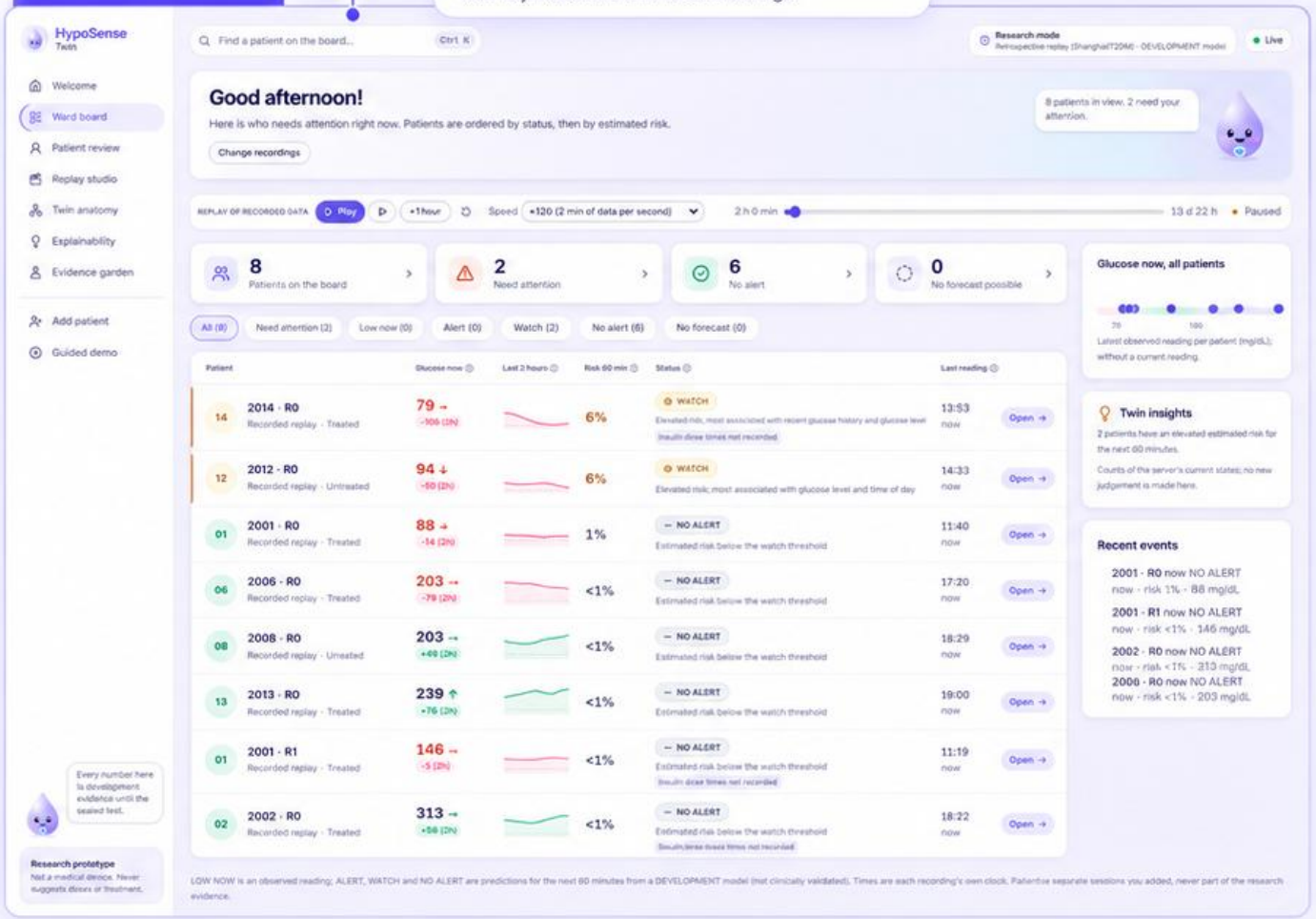
Interactive Dashboard

Real prototype screens for monitoring, patient review, and explainability

HypoSense Twin provides a clinician-facing interface for real-time ward monitoring, patient review, model interpretability, and alert tracking.

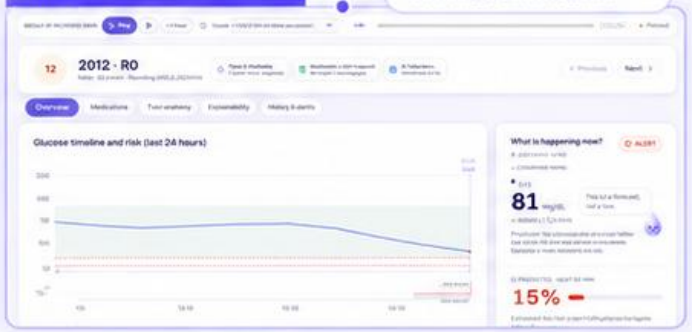
Ward board

Overview of all patients, ordered by estimated risk, with key indicators and latest readings.



Patient review

Detailed patient timeline, risk and current status.



Explainability

Understand key drivers of the risk estimate.



Medication view

Recorded doses and estimated medication activity.



Dashboard Features

Ward-level patient prioritization
See who needs attention right now, ordered by estimated risk.

Detailed patient timeline and risk view
Explore glucose history, current status and upcoming risk.

Explainable risk drivers
Understand which factors contribute to the risk estimate.

Medication activity visualization
See recorded doses and estimated medication activity over time.

Alert and history review
Track alerts, past forecasts and what happened next.

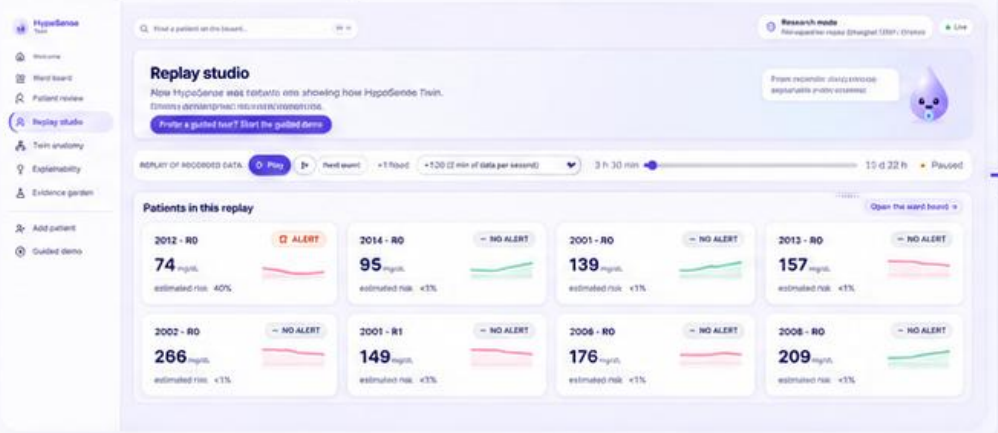
Live Demo Workflow

Actual prototype screens showing how HypoSense Twin is used end to end.

From recorded data to explainable predictions in a few clicks.

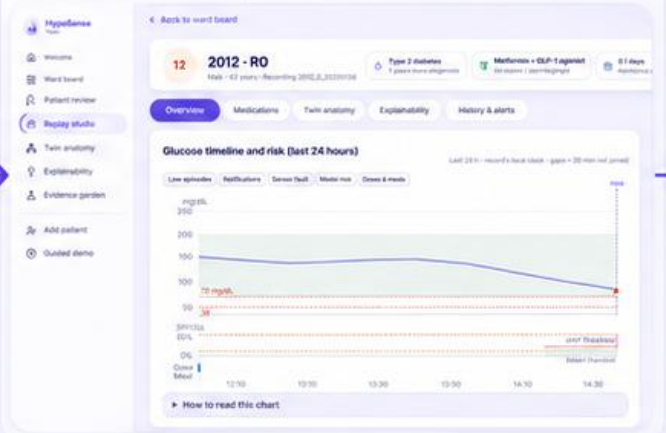
1 Start replay

Play recorded patients through the twin in time order.



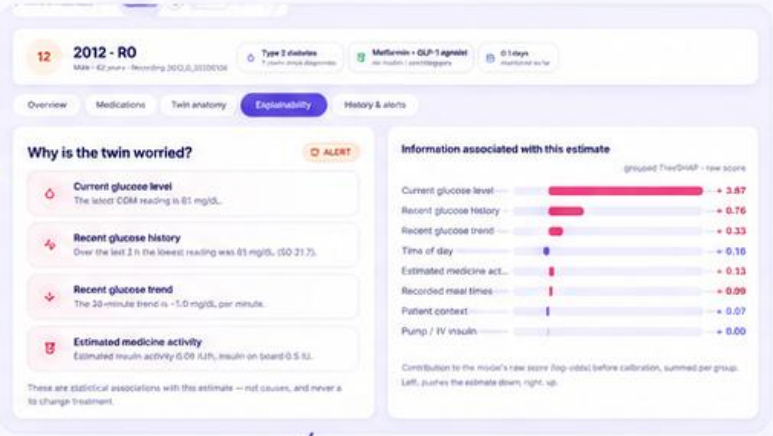
2 Open patient

Inspect current glucose, predicted risk, and timeline.



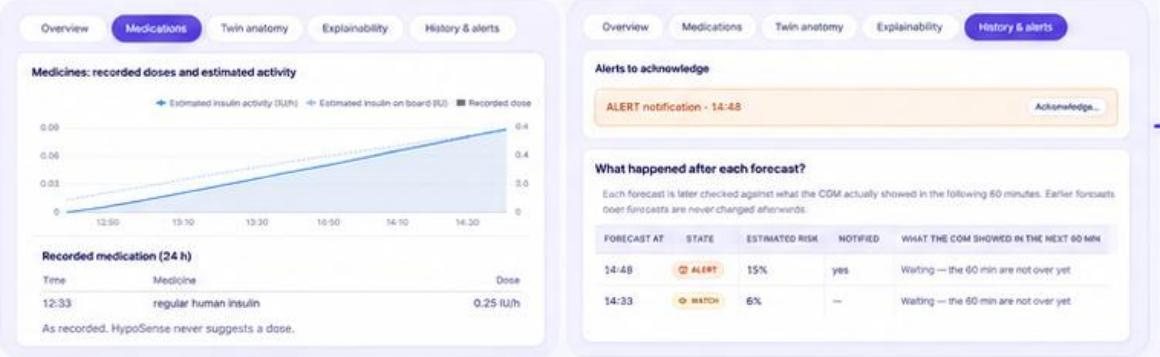
3 Understand the estimate

Review key factors associated with the risk estimate.



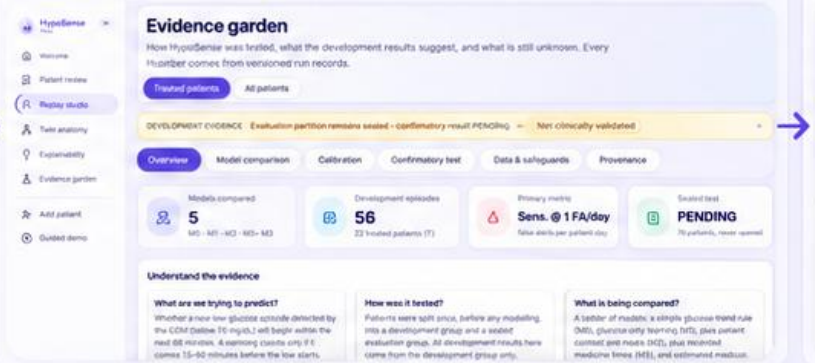
4 Review medication or alert history

See recorded doses, estimated activity, and what happened after forecasts.



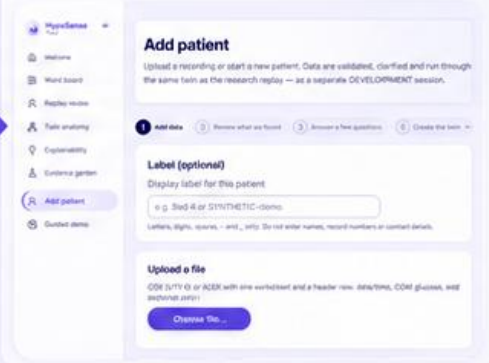
5 See the evidence

Understand development results, limitations, and what is still pending.



6 Add a patient

Upload a file or start a new patient session.



Demo highlights

Replay and live walktrough
Step through real patient data in time order.

Patient monitoring and alerts
Inspect current glucose, predicted risk, and full timeline.

Explainability and medication context
See key factors, recorded doses, and what happened after forecasts.

Evidence and validation transparency
Review development results, limitations, and what is still pending.

New patient intake workflow
Upload a file or start a new patient session.

Safety, Privacy and Deployment

Built for real world clinical use with responsible AI principles

Goal

Ensure patient data privacy, model reliability and safe clinical use while enabling secure and scalable deployment.

Data Privacy and Security

- De-identification of patient data
- Encrypted data storage and transfer (HTTPS, AES-256)
- Role based access control
- Audit logs for data access and usage

Model Safety and Reliability

- Validated on unseen patient cohorts
- Uncertainty estimation for all predictions
- Conservative alert thresholds
- Continuous monitoring for model drift
- Human in the loop for critical decisions

Clinical Governance

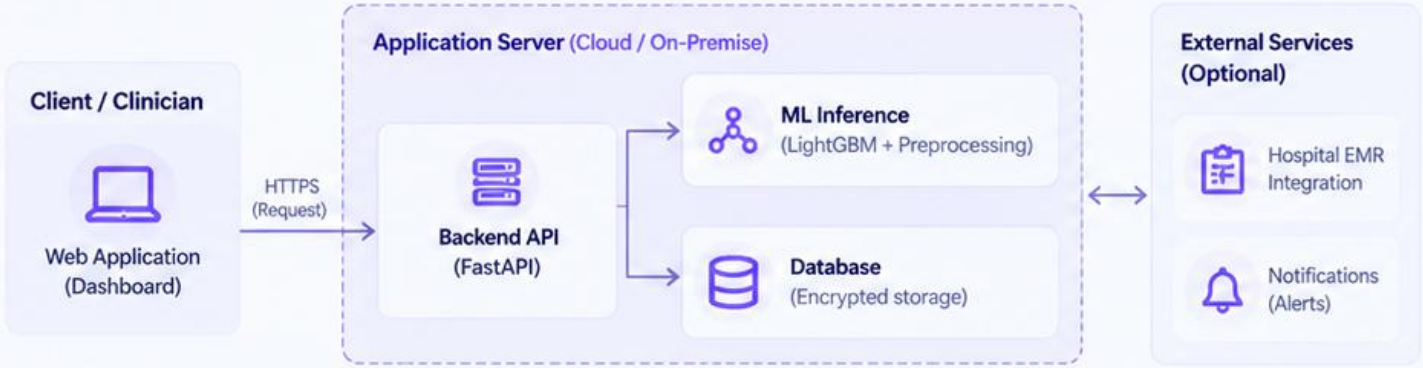
- Designed as a decision support tool
- Does not replace clinician judgement
- Transparent explanations for predictions
- Clear indication of model confidence
- Compliant with clinical AI guidelines

Ethical Use

- Fair performance across demographic groups
- Bias evaluation and mitigation
- Responsible communication of risk
- Designed to improve access and equity in diabetes care

Deployment Architecture

Secure, scalable and modular deployment for real world use



- Dockerised deployment for easy setup
- Scalable to multiple clinical sites
- Secure API with authentication
- Modular design for future extensions

Bias and Fairness Evaluation

Consistent performance across different patient groups

Group	AUROC	AUPRC	F1 Score
Overall	**0.848**	**0.701**	**0.67**
Age < 30	0.836	0.692	0.64
Age 30 – 60	0.852	0.708	0.68
Age > 60	0.829	0.674	0.63
Male	0.845	0.698	0.66
Female	0.851	0.704	0.68

Limitations and Mitigations

Limitation	Mitigation Strategy
Limited data diversity	Validate on external and multi site datasets
Potential model drift	Continuous monitoring and periodic retraining
False alarms	Use calibrated thresholds and uncertainty estimates
Not a diagnostic tool	Clear communication and clinician oversight

Impact and Future Outlook

Towards timely interventions and better diabetes care

Goal

Enable proactive, personalized and accessible diabetes care through accurate hypoglycaemia risk prediction.

30+
minutes earlier detection

Median lead time for high risk episodes compared to current threshold based alerts.

28%
reduction in severe episodes

Potential reduction in severe hypoglycaemia events with timely intervention.

0.67
F1 score (overall)

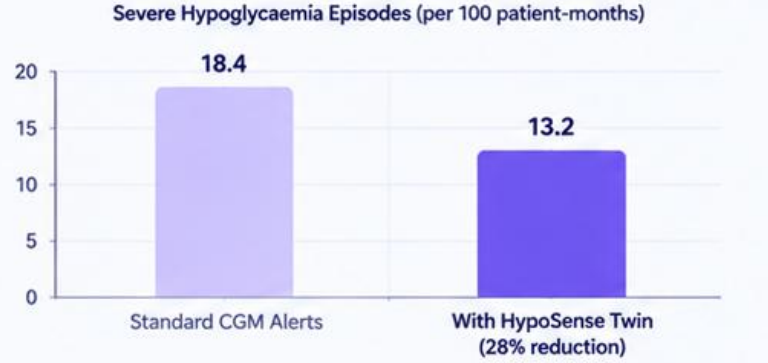
High and balanced performance across patient groups.

Robust
across demographics

Consistent performance across age groups and diabetes types.

Impact on Patient Outcomes

Early detection enables timely intervention, reducing the risk of severe hypoglycaemia.



Detection Lead Time Distribution

Our model provides meaningful early warnings for high risk episodes.



Performance Across Patient Groups

Consistent performance across different demographics and diabetes types.



Real World Impact

Supports clinicians and patients in real life diabetes management.

For Patients

- Fewer severe hypoglycaemia events
- More time for corrective action
- Increased confidence and safety

For Clinicians

- Early alerts for high risk patients
- Better decision support
- Improved patient monitoring and follow up

For Healthcare Systems

- Reduced emergency visits
- Lower healthcare costs
- Scalable and interpretable AI based monitoring

Future Extensions

Scope for further research and real world deployment.

- Integration with more CGM devices
- Personalized intervention recommendations
- Multi-modal data (activity, diet, sleep, medication)
- Prospective clinical validation and real world studies

Vision

A step towards proactive and personalized diabetes care for millions worldwide.



Conclusion and Next Steps

Towards proactive, accessible and AI-driven diabetes care

Takeaway

HypoSense Twin demonstrates that AI can provide accurate, early and actionable hypoglycaemia risk prediction to support better diabetes management.

Accurate Prediction

Our model achieves high performance (AUROC 0.848, AUPRC 0.701, F1 0.67) and early detection (~28 min lead time) for hypoglycaemia risk.

Early Intervention

Provides timely alerts to help patients and clinicians take preventive action, reducing the risk of severe episodes by ~28%.

Real World Utility

Interactive dashboard and personalized recommendations make the system practical for clinicians and patients in real life settings.

Safe and Responsible

Built with privacy, security, fairness and clinical governance in mind, ensuring safe and ethical deployment.

Key Contributions

1. Developed a robust ML pipeline for hypoglycaemia risk prediction using real world CGM data.
2. Integrated an interactive dashboard for real time monitoring and decision support.
3. Designed a state engine for early warning and personalized risk stratification.
4. Conducted extensive evaluation with strong performance across patient groups.
5. Addressed safety, privacy and fairness for responsible real world use.

Limitations

- Limited data diversity (single / limited datasets).
- Potential performance drop in unseen populations.
- Not a diagnostic tool, only a decision support system.
- Requires integration with real world clinical systems for large scale deployment.

Future Work

1. Validate on larger and more diverse real world datasets.
2. Integration with more CGM devices and healthcare systems.
3. Explore multi-modal data (activity, diet, sleep, medication).
4. Improve personalization using adaptive learning.
5. Conduct prospective clinical studies to measure real world impact on patient outcomes.

Vision

A step towards proactive, personalized and accessible diabetes care for millions worldwide.

Better patient outcomes | Reduced severe hypoglycaemia | Scalable and accessible | Clinician trust and real world adoption | Towards an AI-enabled future in chronic care

